TCP Flow & Congestion Control Architecture

Coverage: Sliding Window Flow Control, Silly Window Syndrome, Slow Start, AIMD, Fast Retransmit, TCP Tahoe vs Reno

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1. Flow Control

- Receiver advertises rwnd buffer. Effective Window = min(cwnd, rwnd).
- Silly Window Syndrome solved via Nagle's Algorithm (sender) and Clark's Solution (receiver).

2. Congestion Control

- Slow Start (doubles cwnd per RTT) -> Congestion Avoidance AIMD (+1 MSS per RTT).
- Fast Retransmit & Fast Recovery on 3 duplicate ACKs (TCP Reno).

■ Worked Problem: TCP Reno cwnd Evolution

- Given ssthresh = 16 MSS. Starts at RTT 1 with cwnd = 1 MSS.
- RTT 1: 1 -> RTT 2: 2 -> RTT 3: 4 -> RTT 4: 8 -> RTT 5: 16 (ssthresh reached).
- RTT 6: 17 -> RTT 7: 18 -> RTT 8: 19 MSS.
- 3 Dup ACKs at RTT 8 -> new ssthresh = 9 MSS. Under Reno Fast Recovery: RTT 9 cwnd = 9 MSS, RTT 10 cwnd = 10 MSS.

■ KTU University High-Yield Exam Questions

- Q: Compare TCP Tahoe and TCP Reno congestion control algorithms. (8 Marks)